Acoustic Cues and Listener Variability in Gender-Related Voice Perception

ELEC5305 Final Project Report

Meiqi Song | SID 550200770

Abstract

This project examines how pitch, vocal-tract cues and listener group relate to gender-related voice perception. It reanalyses 240 speech recordings from 20 talkers using the data of Munson and Dolquist (2025). The analysis compares five acoustic conditions, examines response concentration and predicts the group-averaged percentage of "man" responses. Within-talker comparisons showed a decrease of 23.73 percentage points for raised formants, with a 95% bootstrap confidence interval of [−31.06, −16.60]. Cue-consistent recordings had higher response concentration than cue-conflict recordings. The mean within-talker difference was 5.78 points [0.99, 10.65]. In five-fold validation with separate talkers in training and testing, F0 and acoustic vocal-tract length together achieved $R^2 = 0.797$ and RMSE = 15.78 points. Adding MFCC features did not improve performance. These results support a combined role for acoustic cues, but responses also varied across listener groups. The findings describe listener judgments and do not establish a speaker's gender identity.

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Project site: https://meiqi0747.github.io/elec5305-project-550200770/

Code repository: https://github.com/meiqi0747/elec5305-project-550200770

GitHub username: meiqi0747

Demonstration video: https://youtu.be/cVo23rYBB2s

1 Introduction

Listeners use pitch and resonance to form impressions of gender, but the same voice can receive different judgments. The central research question is: To what extent can F0 and vocal-tract cues explain gender-related voice judgments across acoustic conditions and listener groups? This project addresses it through condition comparisons, response concentration and prediction for unseen talkers. It uses published perception data and a MATLAB feature-analysis pipeline.

The proposal planned binary speaker classification with KNN and SVM. The final project instead predicts listener response percentages. This target better fits the selected dataset, which records varied judgments rather than a binary classification target. The project retains the proposal's focus on acoustic feature comparison and separation of talkers in training and testing.

2 Literature review

2.1 Pitch and resonance cues

Fundamental frequency (F0) describes the vocal source. Formants describe vocal-tract resonances. Lower formants generally correspond to longer acoustic vocal-tract length (aVTL). This estimate is not a direct anatomical measurement. The distinction motivates testing F0 and aVTL separately and together.

Skuk and Schweinberger (2014) manipulated F0, formants, aperiodicity and spectrum level in voice morphs. F0 was the strongest single cue in their task. Formants also contributed, and changing F0 and formants together produced strong changes in perceived gender. Their results support a model with both source and resonance information. However, the experiment used controlled syllable morphs. Its relative cue weights should not be assumed to hold for all sentences or talkers.

Pernet and Belin (2012) also examined pitch and timbre through manipulated voices. They found that listeners could categorise voices using pitch alone, but pitch was especially useful when timbre was ambiguous. This suggests that cue use depends on the available information. It also explains why a single fixed threshold for F0 would be an incomplete account of voice perception. The present project tests the practical value of combining two interpretable cues without assuming that one cue always dominates.

2.2 Listener identity and response choices

Hope and Lilley (2022) studied ambiguous synthetic voices using separate masculinity, femininity and "other" scales. Gender-expansive listeners used the scales more broadly and systematically categorised some neutral voices as nonbinary. Their work shows that the listener's conception of gender can influence how an acoustic signal is labelled. A lower rate of one response therefore need not indicate poorer hearing or less accurate perception.

Munson and Dolquist (2025) provide the data used here. They examined sentence recordings from the Palette of Voices with cisgender heterosexual women, cisgender heterosexual men, and gender and sexuality expansive listeners. Their analyses showed differences in acoustic cue weighting between groups. They also distinguished gender categorisation from judgments about whether a person was cisgender or transgender. This project focuses on the first task. It does not model perceived trans status.

2.3 Contribution of this project

This project compares F0, aVTL and MFCC features under talker-separated validation. It also tests response concentration within talkers. The grouped table supports descriptive listener-group comparisons, but cannot reproduce individual-listener modelling. Whether richer spectral features improve prediction remains an empirical question.

3 Methodology

3.1 Data and source experiment

The data comprise 240 stimulus records, 240 matching WAV files and 20 talkers. There are 48 records in each of five acoustic conditions. All WAV files use a 44.1 kHz sample rate. Their durations range from 1.73 to 6.27 seconds. The acoustic and perception measurements come from MunsonDolquist2025_SummaryPerceptionData.xlsx. The MATLAB analysis joins each record to its audio file using the filename.

In the source experiment, 199 listeners completed an online task: 62 cisgender heterosexual women (CHF), 87 cisgender heterosexual men (CHM), and 50 gender and sexuality expansive listeners (GSE). Each listener heard one set of 60 recordings. The source corpus includes transgender and nonbinary talkers, cisgender men, and talkers who described themselves as male without further qualification (Munson & Dolquist, 2025). These source-study participants are not new participants recruited for this project.

The original study used Praat to raise or lower F0 or formant frequencies by 10%. Filename suffixes identify the five stimulus types: X is original, A is lowered F0, Y is raised F0, Z is lowered formants, and B is raised formants. These suffixes should not be confused with the source experiment's four presentation lists. Comparisons here use the available recordings in each stimulus type. They are not comparisons of the same sentence before and after processing.

3.2 Acoustic and response measures

The published F0 value is the mean of measurements taken at the midpoint of primary- and secondary-stressed vowels in a sentence. The published aVTL value is calculated from F1 to F4 using an odd-quarter tube approximation, then averaged over vowels. This project uses those published measures. It does not claim to have remeasured F0 or formants from the WAV files.

For each recording, the percentages for CHF, CHM and GSE are averaged with equal group weights. The main target is the resulting percentage of "man" responses. Equal weighting gives each listener group the same contribution, despite their different sample sizes. It is not a pooled percentage across all 199 listeners.

Response concentration is the largest of the four group-averaged percentages: man, woman, other gender, and "no" to being reasonably certain about guessing gender. The analysis code calls this ListenerAgreement. This report uses response concentration because a high value can reflect shared uncertainty. It is neither a measure of correctness nor a reliability coefficient. The four percentages can sum to slightly above or below 100 because the source values are rounded.

3.3 Statistical comparisons

Condition means first describe the recordings as a whole. For the main comparisons, each talker's mean in a manipulated condition is compared with that talker's original-condition mean. The 20 talker differences are then averaged equally. This reduces the influence of talkers with more recordings in a condition, but it does not remove sentence-content differences.

The final confidence intervals use 3,000 bootstrap resamples of talker differences with replacement. The MATLAB seed is 5305. The 2.5th and 97.5th percentiles are obtained by interpolation. These are unadjusted 95% intervals. They represent uncertainty across the sampled talkers, not uncertainty across independently resampled listeners.

Cue consistency is defined using the full dataset. F0 and aVTL are separately standardised. The two directional scores are $-z(F0)$ and $z(aVTL)$. A recording is cue-consistent when their product is zero or positive, and cue-conflict when it is negative. Thus, low F0 with high aVTL and high F0 with low aVTL both count as consistent. This is a dataset-relative rule, not a biological gender boundary. There are 162 consistent and 78 conflicting records.

A first analysis compares pooled cue-group means using 2,000 talker-cluster bootstrap resamples. The stronger within-talker check compares the two cue-group means separately for each talker, then averages their differences. Nineteen talkers have both cue types. Talker 15 is excluded only from this comparison. The final within-talker cue interval uses 3,000 resamples.

3.4 MFCC extraction and prediction

Mel-frequency cepstral coefficients (MFCCs) summarise short-time spectral shape. The MATLAB pipeline converts multichannel audio to mono, removes the mean and normalises peak amplitude. It applies pre-emphasis with coefficient 0.97. The signal is split into 25 ms Hamming-windowed frames with a 10 ms step. A power spectrum is passed through 26 triangular mel filters. Log filter energies are transformed using a discrete cosine transform. Coefficients 1 to 13 are retained, excluding the zeroth coefficient. Their means and standard deviations give 26 features per recording.

Four ridge-regression models predict the group-averaged "man" response: F0 only, aVTL only, F0 plus aVTL, and F0 plus aVTL plus 26 MFCC features. Five folds are assigned cyclically by sorted talker ID. Each fold holds out four entire talkers. Features are standardised using training-fold statistics only. The ridge penalty is fixed at 1, and the intercept is not penalised. Performance is calculated across all held-out predictions. Root mean square error (RMSE) is measured in percentage points. The coefficient of determination (R^2) compares squared prediction errors with variation around the overall observed mean.

Ridge regression suits the continuous response target and limits large coefficients when predictors overlap. It is simpler to inspect than a large neural model, but cannot capture complex nonlinear relationships. The main technical distinctions are source versus filter cues, group averages versus individual responses, and stimulus-level versus talker-level independence. The validation and bootstrap designs address the last distinction in different ways.

4 Results

4.1 Acoustic conditions

Figure 1 shows substantial overlap between conditions. Raised formants tend to have shorter aVTL values, while lowered formants tend to have longer values. Table 1 gives the descriptive means. The observed shifts are not expected to equal the nominal 10% manipulation, because the condition groups contain different sentence recordings.

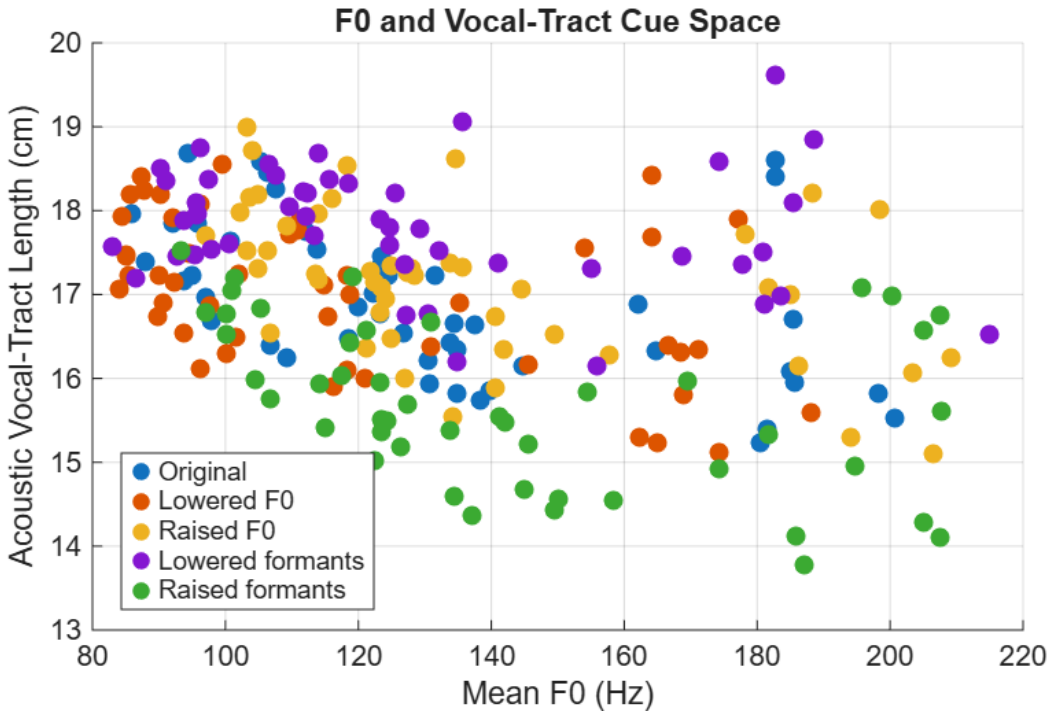


Figure 1. F0 and acoustic vocal-tract length for all 240 recordings. Each point is one stimulus, coloured by condition. aVTL is an acoustic estimate in centimetres.

Table 1. Descriptive condition means, with 48 recordings per condition.

Condition	F0 (Hz)	aVTL (cm)	Man (%)	Concentration (%)
Original	130.82	16.92	63.00	76.49
Lowered F0	118.80	17.01	63.36	76.37
Raised F0	136.70	17.15	58.55	74.66
Lowered formants	127.86	17.81	73.30	82.59
Raised formants	142.56	15.69	38.50	64.09

The lowered-formant condition has the highest mean “man” response, at 73.30%. The raised-formant condition has the lowest, at 38.50%. The original condition is 63.00%. Lowering F0 gives a similar pooled mean to the original condition. Raising F0 gives a lower mean. These pooled results provide context; the within-talker estimates below are the main comparisons.

4.2 Listener groups and man responses

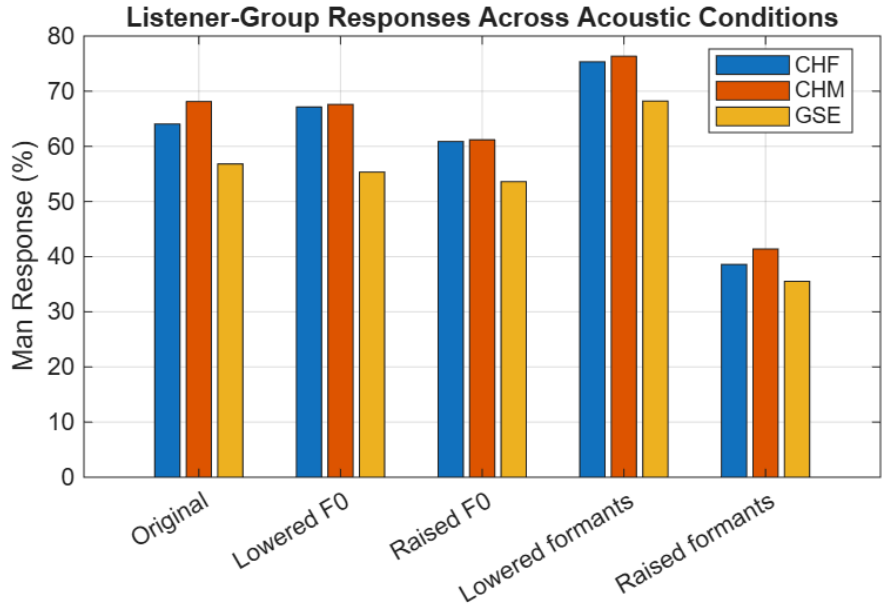


Figure 3. Mean "man" response by listener group and condition. Each bar averages 48 recordings. Group abbreviations are defined in Section 3.1.

GSE listeners give fewer "man" responses in every condition. For original recordings, the means are 56.81% for GSE, 64.05% for CHF and 68.13% for CHM. These are descriptive group differences, not individual-listener effects or accuracy differences.

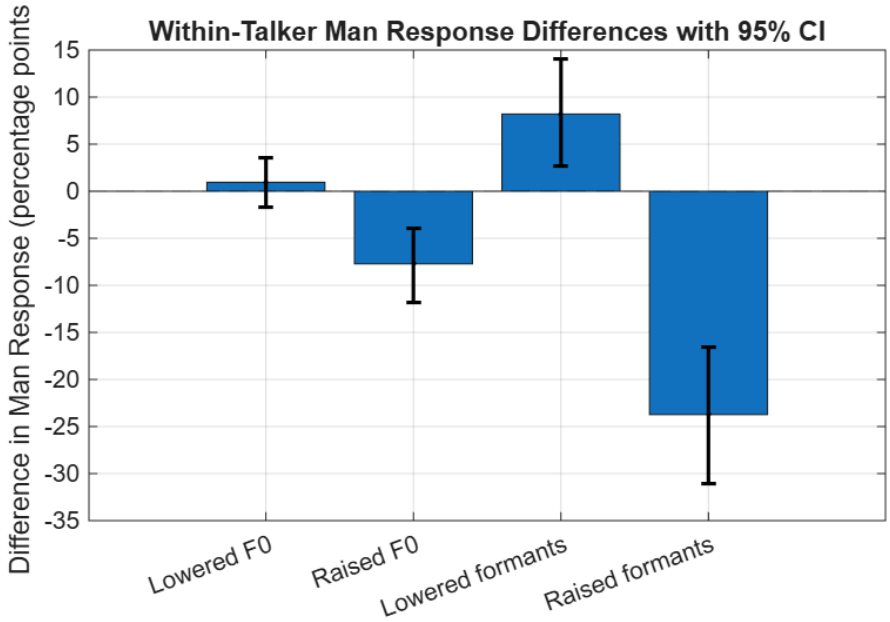


Figure 12. Within-talker changes in "man" response relative to original recordings. Error bars are 95% bootstrap intervals from 3,000 resamples of 20 talkers.

Lowered F0 gives +0.94 points [−1.70, 3.58], raised F0 −7.72 [−11.84, −3.99], lowered formants +8.20 [2.72, 14.07], and raised formants −23.73 [−31.06, −16.60]. Raised formants also differ by +9.64 Hz in F0 and −1.28 cm in aVTL within talkers. Their response difference cannot be attributed to formants alone.

4.3 Concentration and cue consistency

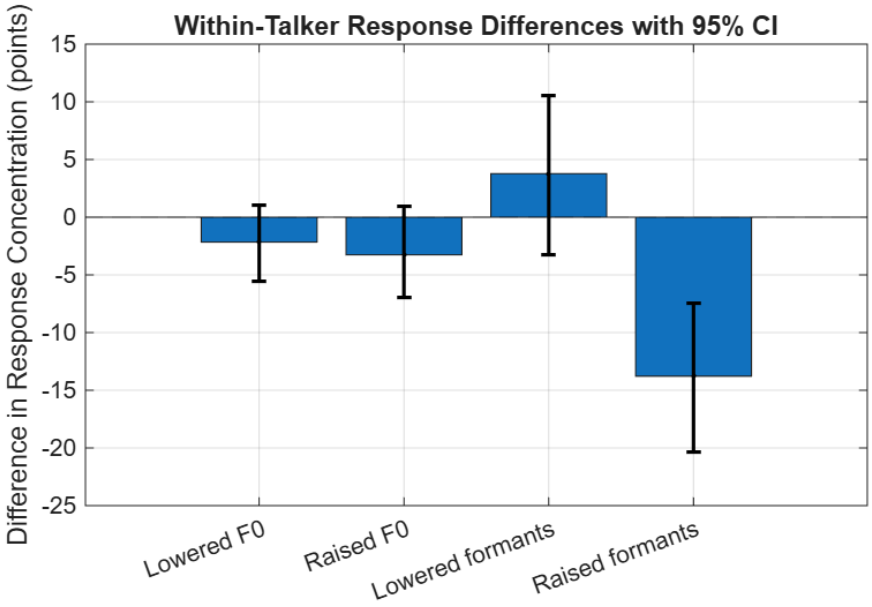


Figure 13. Within-talker concentration changes relative to original recordings, with 95% bootstrap intervals. Concentration is the largest of four group-averaged response percentages.

Raised formants reduce concentration by 13.78 points [−20.31, −7.49]. Lowered F0 (−2.16), raised F0 (−3.26) and lowered formants (+3.78) have intervals containing zero. A shift in “man” responses does not always imply a clear change in concentration.

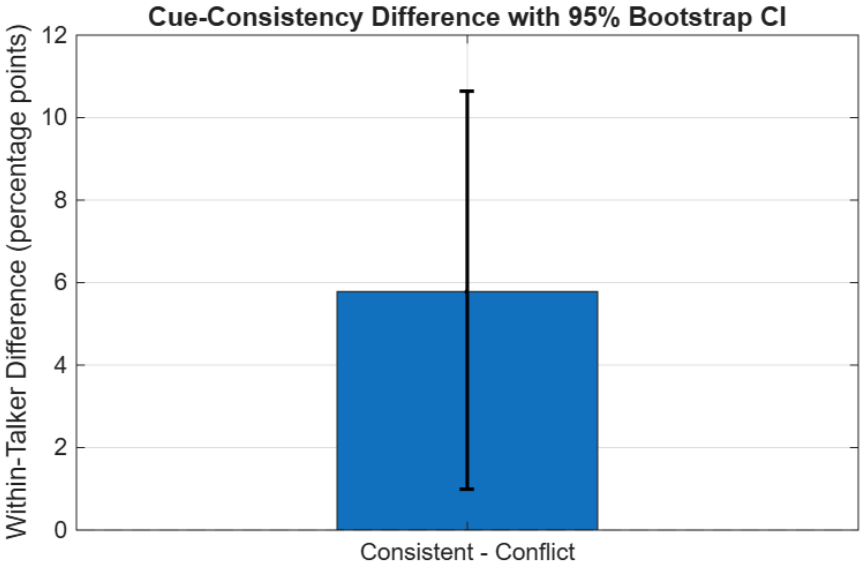


Figure 14. Consistent-minus-conflict concentration difference within 19 talkers: 5.78 points, 95% bootstrap CI [0.99, 10.65].

Pooled concentration is 79.09% for consistent recordings and 66.01% for conflicting recordings: a difference of 13.09 points [5.07, 20.96]. The within-talker estimate is smaller. Fourteen of 19 talkers have positive differences, ranging overall from −13.71 to +25.84 points. The association is not universal, and the pooled difference partly reflects talker composition. Sentence differences remain in both analyses.

4.4 Prediction for unseen talkers

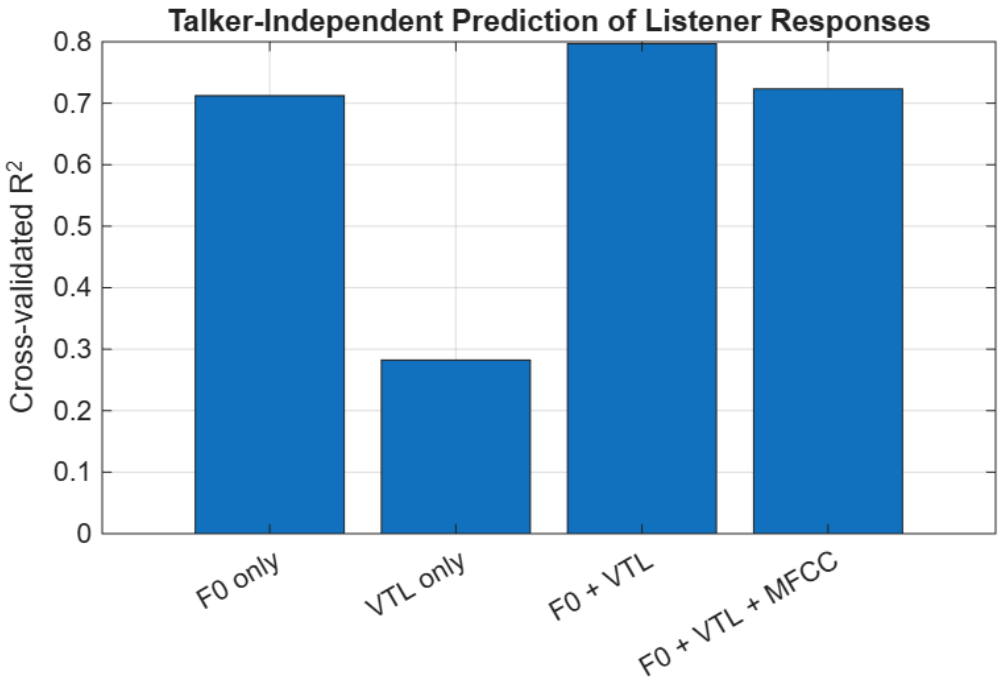


Figure 5. Cross-validated R^2 for four ridge-regression feature sets. All recordings from a talker remain in one fold. "VTL" in the figure refers to acoustic vocal-tract length. The MFCC model adds 26 spectral summary features.

Table 2. Performance across held-out predictions for all 240 recordings.

Feature set	RMSE (points)	R^2
F0 only	18.77	0.712
VTL only	29.65	0.282
F0 + VTL	15.78	0.797
F0 + VTL + MFCC	18.41	0.723

F0 alone predicts responses better than aVTL alone. Combining them improves RMSE from 18.77 to 15.78 points and raises R^2 from 0.712 to 0.797. The two cues therefore provide useful complementary information under this validation split.

Adding MFCCs increases RMSE to 18.41 points and reduces R^2 to 0.723. More features did not improve generalisation here. Possible reasons include the small number of talkers, redundant spectral information and the fixed penalty. These results do not show that MFCCs are generally unhelpful. The model scores have no uncertainty intervals, so their ranking should be treated as a result for this evaluation rather than a proven difference across populations.

5 Discussion and limitations

The prediction results support using F0 and aVTL together. This agrees with the broader evidence that both source and resonance cues matter in gender-related voice perception (Skuk & Schweinberger, 2014). It also fits Pernet and Belin's (2012) finding that the use of pitch can depend on timbre. The present models are additive, so they do not directly test perceptual cue interactions.

The large raised-formant response difference should be interpreted alongside the prediction results. It does not contradict the stronger performance of F0 alone. A condition comparison measures differences between particular stimulus sets. Prediction measures how well features explain responses across the whole sample. The conditions also differ in measured F0 and sentence content. These analyses cannot establish a universal ranking of cue importance.

The three listener groups give different average responses to the same stimulus sets. This is consistent with Hope and Lilley (2022) and Munson and Dolquist (2025). The source study also reports variation within groups. Our group averages cannot show how much individual listeners differ, so they should not be used to describe every listener in the same way.

Several limitations affect the conclusions. First, the sample contains only 20 talkers from one corpus. It does not represent all gender identities, languages or speaking styles. Second, condition comparisons are paired by talker but not by identical utterance. Sentence content and other acoustic cues can still contribute to the estimated differences.

The response data also have limits. They are rounded group percentages, and equal group weighting differs from pooling individual listeners. The bootstrap resamples talkers only. It does not capture listener-level sampling uncertainty, and the intervals are not adjusted for multiple comparisons. An interval excluding zero is therefore evidence within this analysis, rather than proof of a general effect.

Response concentration combines different outcomes. A high value may mean that listeners agree on "man", or that many are not confident enough to guess. Analysing uncertainty separately would make this distinction clearer. The cue-consistency threshold also depends on sample means. Testing other thresholds and continuous cue interactions would show how much the result depends on this choice.

Finally, the models use one deterministic five-fold split and a fixed ridge penalty. Linear predictions are not constrained to the 0–100 range. Future evaluation could use repeated talker-grouped splits and tune the penalty within training folds. Individual response data would allow models with both listener and talker effects. Such extensions would help separate acoustic associations from differences in response strategy.

6 Conclusion

F0 and aVTL together gave the best prediction of group-averaged "man" responses in this project. Raised formants showed the largest within-talker decrease in both "man" responses and response concentration. Cue-consistent recordings had higher concentration on average, but the difference was smaller within talkers than in pooled data. These findings describe patterns in listener judgments. They should not be used to determine a person's identity.

The project answers the acoustic prediction part of the research question more directly than the listener-variation part. Group differences remain descriptive. Individual responses and a broader talker sample are needed for a fuller explanation.

Academic integrity and source acknowledgement

This report is a secondary analysis of the data and audio associated with Munson and Dolquist (2025). The original authors collected the recordings, listener responses and published acoustic measurements. No new participants were recruited for this project. The project MATLAB analysis supplies the MFCC features, descriptive comparisons, predictive models and bootstrap summaries.

Figures 1, 3, 5, 12, 13 and 14 are project analysis figures generated from these data. Their original numbers are retained to match the MATLAB outputs. Numerical results are taken from the project CSV tables. Condition means and model scores were independently recomputed from the source workbook and saved feature table. Bootstrap interval endpoints were checked against the exported tables and the documented MATLAB procedure.

ChatGPT/Codex assisted with report organisation, numerical checks and document formatting.

References

Hope, M., & Lilley, J. (2022). Gender expansive listeners utilize a non-binary, multidimensional conception of gender to inform voice gender perception. Brain and Language, 224, 105049. https://doi.org/10.1016/j.bandl.2021.105049

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Data source

Munson and Dolquist (2025), supplementary perception data and speech stimuli: https://osf.io/n3twm/. Local analysis inputs are the summary perception workbook and the corresponding 240 WAV files. The numerical exports are Final_Analysis_Table.csv, Final_Model_Results.csv and the tables in Final_Results.